

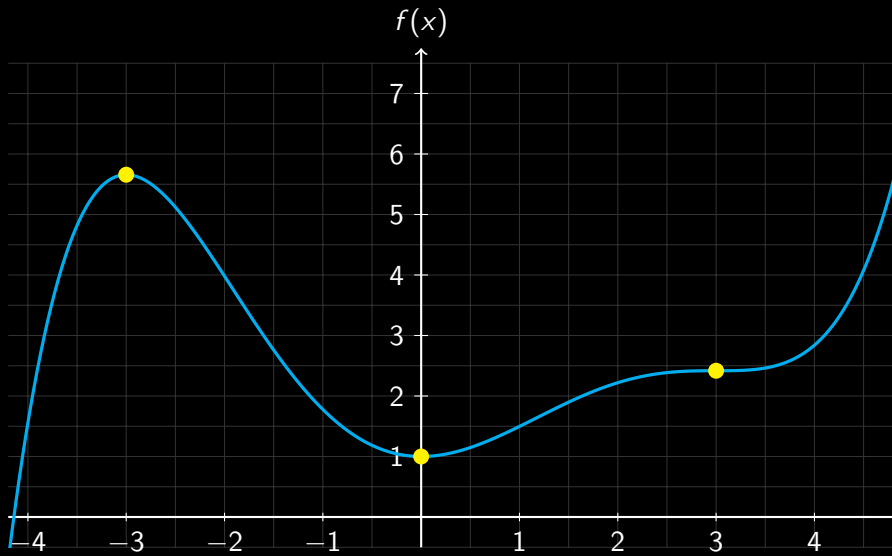
1202SCG Calculus 1
1020ENG Engineering Mathematics 1

Week 5 Lecture

David Harman

Applications of Derivatives

- ▶ Locate where a function changes direction.
- ▶ Classify stationary points using signs or curvature.
- ▶ Describe graph shape using concavity and inflection points.
- ▶ Optimise a quantity subject to constraints.
- ▶ Estimate the effect of a small input change.



Stationary Points

A **stationary point** occurs at $x = a$ when

$$f'(a) = 0.$$

The tangent to the curve is horizontal at this point.

A stationary point may be a

- ▶ local maximum,
- ▶ local minimum, or
- ▶ stationary point of inflection.

Finding Stationary Points

To find the stationary points of $y = f(x)$:

1. Differentiate to find $f'(x)$.
2. Solve $f'(x) = 0$.
3. Substitute each solution into $f(x)$.
4. Report each point as a coordinate $(x, f(x))$.

Remember to find the y -coordinate.

Stationary Points: Example

Find the stationary points of

$$f(x) = x^3 - 3x^2 - 9x + 5.$$

$$\begin{aligned} f'(x) &= 3x^2 - 6x - 9 \\ &= 3(x + 1)(x - 3). \end{aligned}$$

Hence $f'(x) = 0$ when

$$x = -1 \quad \text{or} \quad x = 3.$$

Since $f(-1) = 10$ and $f(3) = -22$, the stationary points are

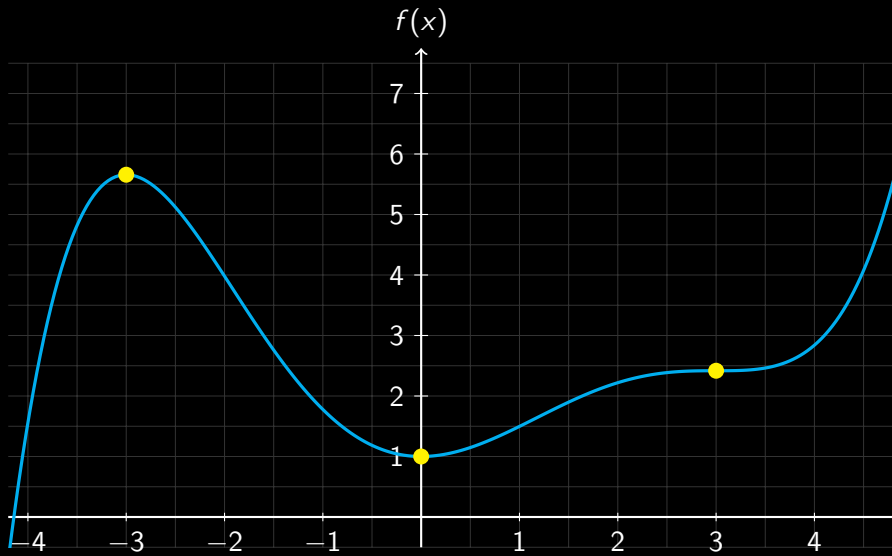
$$(-1, 10) \quad \text{and} \quad (3, -22).$$

First-Derivative Test

The sign of the first derivative determines if the function is increasing or decreasing at that point.

Sign of $f'(x)$	Behaviour of $f(x)$
$f'(x) > 0$	Increasing
$f'(x) < 0$	Decreasing

Examine the sign of $f'(x)$ immediately before and after each stationary value.



Classifying Stationary Points

At a stationary value $x = a$:

Change in $f'(x)$	Classification
Positive to negative	Local maximum
Negative to positive	Local minimum
No change of sign	Neither

First-Derivative Test: Example

For

$$f'(x) = 3(x + 1)(x - 3),$$

test the intervals separated by -1 and 3 .

increasing
 $f'(x) > 0$

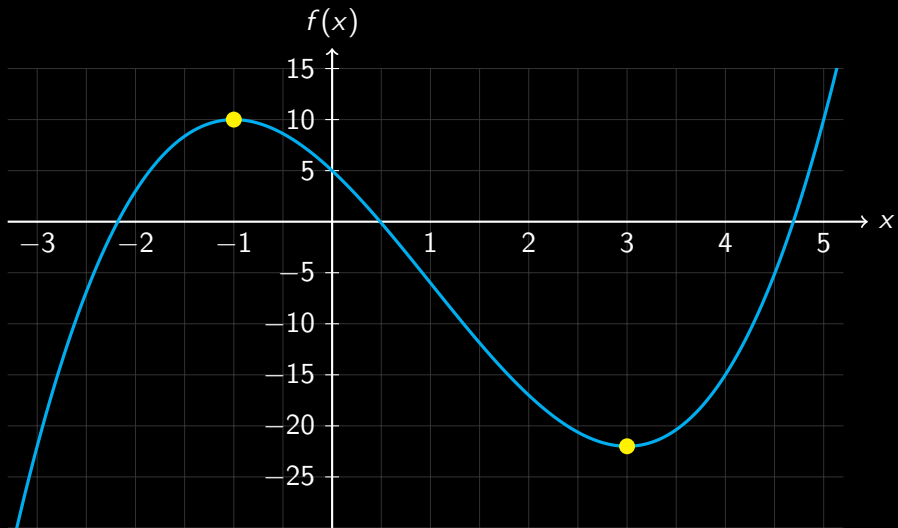
decreasing
 $f'(x) < 0$

increasing
 $f'(x) > 0$

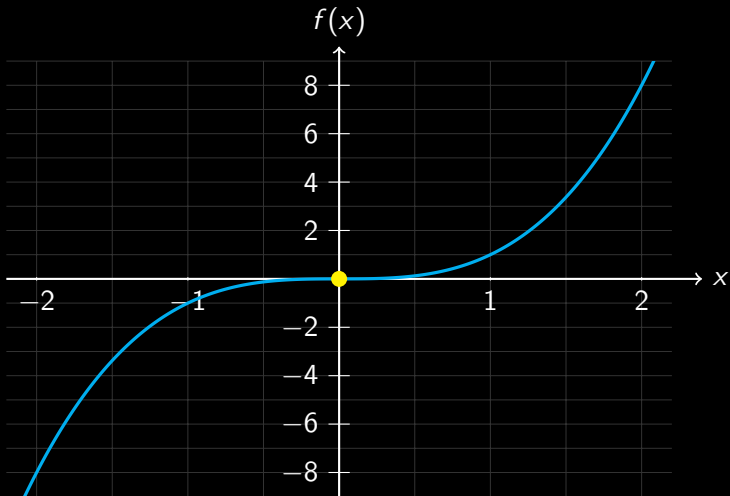


- ▶ At $x = -1$: increasing to decreasing, so $(-1, 10)$ is a local maximum.
- ▶ At $x = 3$: decreasing to increasing, so $(3, -22)$ is a local minimum.

$$f(x) = x^3 - 3x^2 - 9x + 5$$



A Stationary Point May Not Be A Turning Point



A Stationary Point May Not Be A Turning Point

Consider

$$f(x) = x^3, \quad f'(x) = 3x^2.$$

Since $f'(0) = 0$, $(0, 0)$ is a stationary point. However,

$$f'(x) > 0 \quad \text{on both sides of } x = 0.$$

Therefore $(0, 0)$ is not a turning point.

First-Derivative Test: Practice

Find and classify every stationary point.

1. $f(x) = x^2 - 6x + 5$

2. $g(x) = x^3 - 12x$

3. $h(x) = x^4$

Use a sign chart for each derivative.

First-Derivative Test: Answers

1. $f'(x) = 2x - 6$: local minimum at $(3, -4)$.
2. $g'(x) = 3(x - 2)(x + 2)$: local maximum at $(-2, 16)$ and local minimum at $(2, -16)$.
3. $h'(x) = 4x^3$: local minimum at $(0, 0)$.

Concavity

Concavity describes how the gradient changes as x increases.

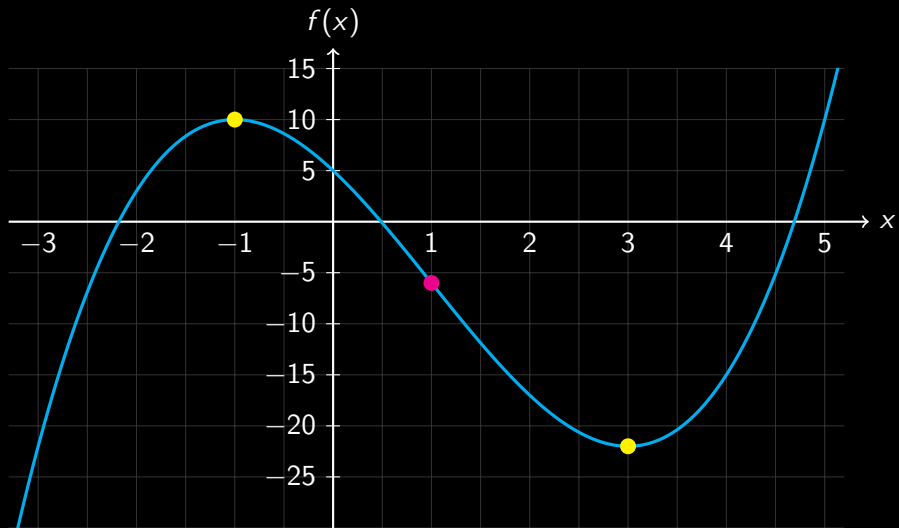
Concave up $\cup$

$f'(x)$ is increasing,
 $f''(x) > 0$.

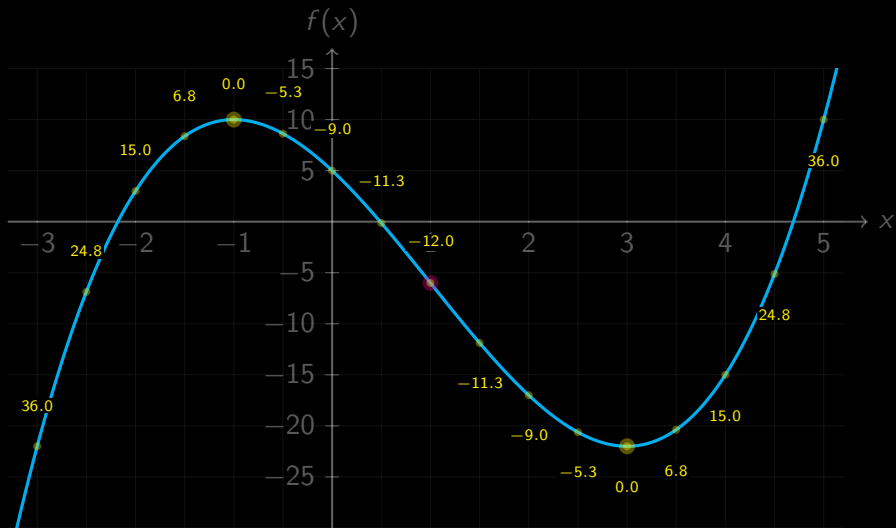
Concave down $\cap$

$f'(x)$ is decreasing,
 $f''(x) < 0$.

$$f(x) = x^3 - 3x^2 - 9x + 5$$



$$f(x) = x^3 - 3x^2 - 9x + 5$$



Finding Intervals of Concavity

For

$$f(x) = x^3 - 3x^2 - 9x + 5,$$

we have

$$f'(x) = 3x^2 - 6x - 9,$$

$$f''(x) = 6x - 6.$$

Therefore,

- ▶ $f''(x) < 0$ when $x < 1$, so the curve is concave down;
- ▶ $f''(x) > 0$ when $x > 1$, so the curve is concave up.

Points of Inflection

A **point of inflection** occurs where the concavity changes.

Possible points of inflection occur where

$$f''(x) = 0$$

or where $f''(x)$ is undefined.

It is only a point of inflection if the sign of $f''(x)$ changes.

Point of Inflection: Example

For $f(x) = x^3 - 3x^2 - 9x + 5$,

$$f''(x) = 6x - 6,$$

$$0 = 6x - 6,$$

$$x = 1.$$

The second derivative changes from negative to positive at $x = 1$, so the concavity changes from down to up.

Since

$$f(1) = 1 - 3 - 9 + 5 = -6,$$

the point of inflection is $(1, -6)$.

Point of Inflection: Another Example

Find the point of inflection of

$$f(x) = x^3 + x.$$

$$f'(x) = 3x^2 + 1,$$

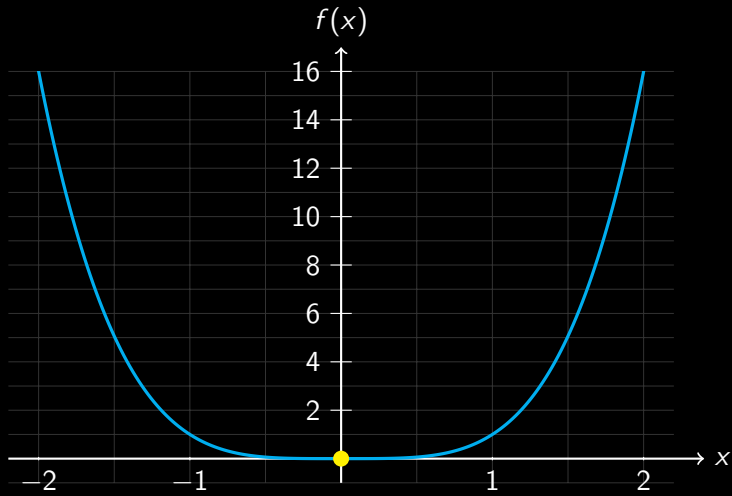
$$f''(x) = 6x.$$

Solving $f''(x) = 0$ gives $x = 0$. The second derivative changes from negative to positive at $x = 0$, so the concavity changes from down to up.

Since $f(0) = 0$, the point of inflection is $(0, 0)$.

This point is not stationary because $f'(0) = 1$.

Zero Second Derivative Is Not Enough



Zero Second Derivative Is Not Enough

Consider

$$f(x) = x^4, \quad f''(x) = 12x^2.$$

Although $f''(0) = 0$,

$$f''(x) > 0 \quad \text{on both sides of } x = 0.$$

The function is concave up on both sides, so the concavity does not change.

Therefore $(0, 0)$ is **not** a point of inflection.

Second-Derivative Test

Suppose $x = a$ is a stationary value, so $f'(a) = 0$.

Value of $f''(a)$	Classification
$f''(a) > 0$	Local minimum
$f''(a) < 0$	Local maximum
$f''(a) = 0$	Test is inconclusive

Second-Derivative Test: Example

Find and classify the stationary points of

$$f(x) = 2x^3 - 3x^2 - 12x + 4.$$

we obtain

$$\begin{aligned}f'(x) &= 6x^2 - 6x - 12 \\ &= 6(x + 1)(x - 2),\end{aligned}$$

$$f'(x) = 0 \qquad \Rightarrow x = -1, 2,$$

$$f''(x) = 12x - 6.$$

Therefore,

$$f''(-1) = -18 < 0 \qquad \Rightarrow \text{local maximum at } (-1, 11),$$

$$f''(2) = 18 > 0 \qquad \Rightarrow \text{local minimum at } (2, -16).$$

When the Second-Derivative Test Fails

For $f(x) = x^4$, the stationary point is $(0, 0)$ and

$$f''(0) = 0.$$

This does not mean there is no maximum or minimum. It means the test gives no answer.

Return to the first-derivative test:

$$f'(x) = 4x^3$$

changes from negative to positive, so $(0, 0)$ is a local minimum.

Choosing a Classification Test

First-derivative test

- ▶ Always decisive when signs can be tested.
- ▶ Shows increasing and decreasing behaviour.
- ▶ Requires intervals on both sides.

Second-derivative test

- ▶ Often quicker when f'' is simple.
- ▶ Shows local curvature.
- ▶ Inconclusive when $f''(a) = 0$.

Complete Curve Analysis

$$f(x) = x^3 - 3x^2 - 9x + 5,$$

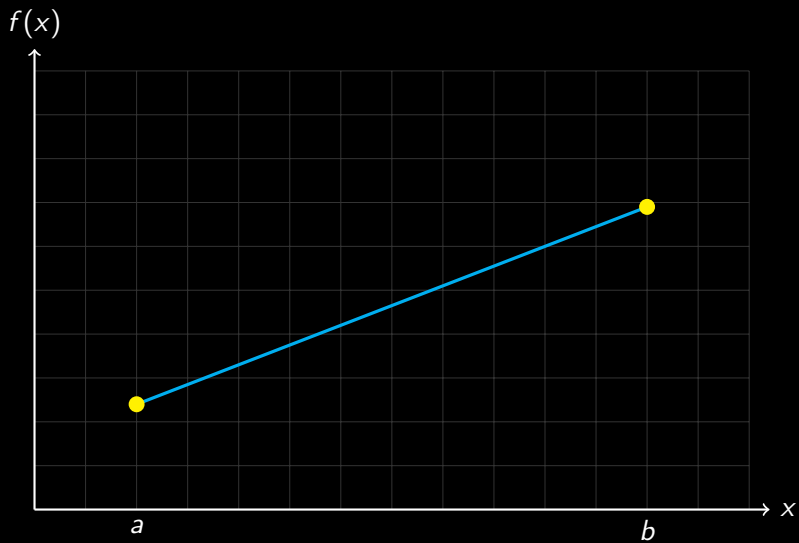
Feature	Result
First derivative	$f'(x) = 3(x + 1)(x - 3)$
Stationary points	Maximum $(-1, 10)$; minimum $(3, -22)$
Increasing	$x < -1$ and $x > 3$
Decreasing	$-1 < x < 3$
Second derivative	$f''(x) = 6x - 6$
Concavity	Down for $x < 1$; up for $x > 1$
Point of inflection	$(1, -6)$

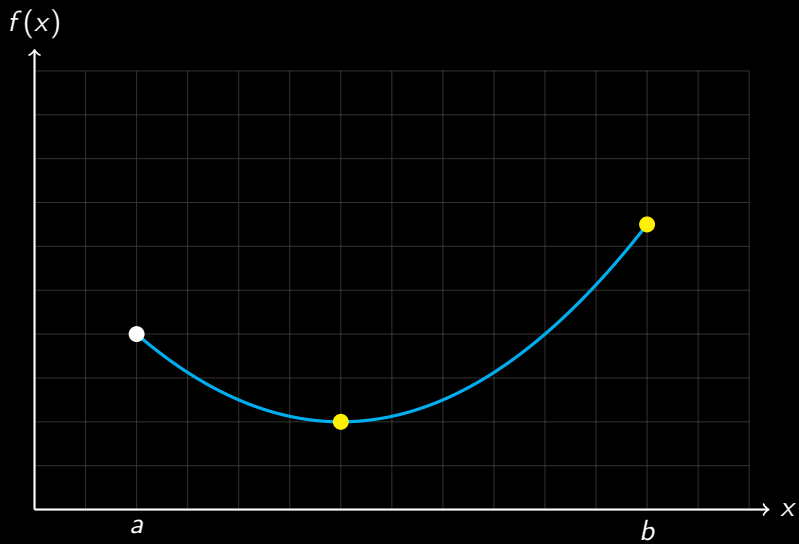
Absolute Extrema on a Closed Interval

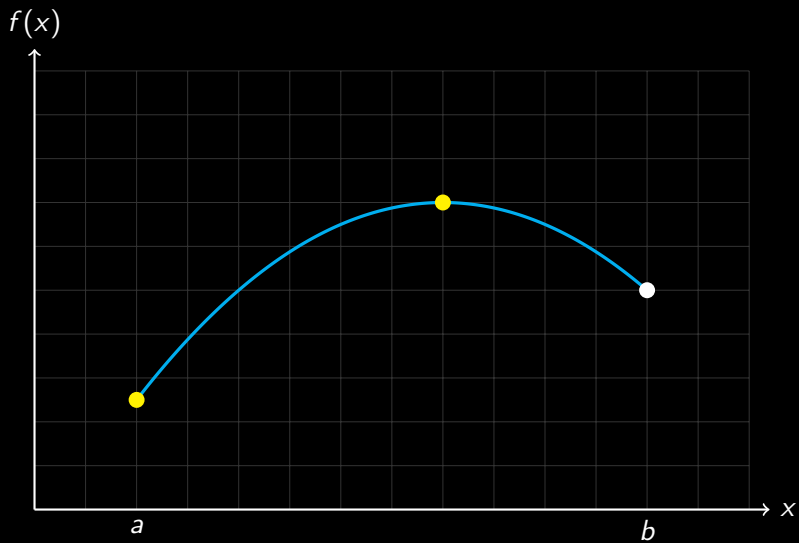
For a continuous function on $[a, b]$, absolute extrema may occur at interior stationary points or at the endpoints.

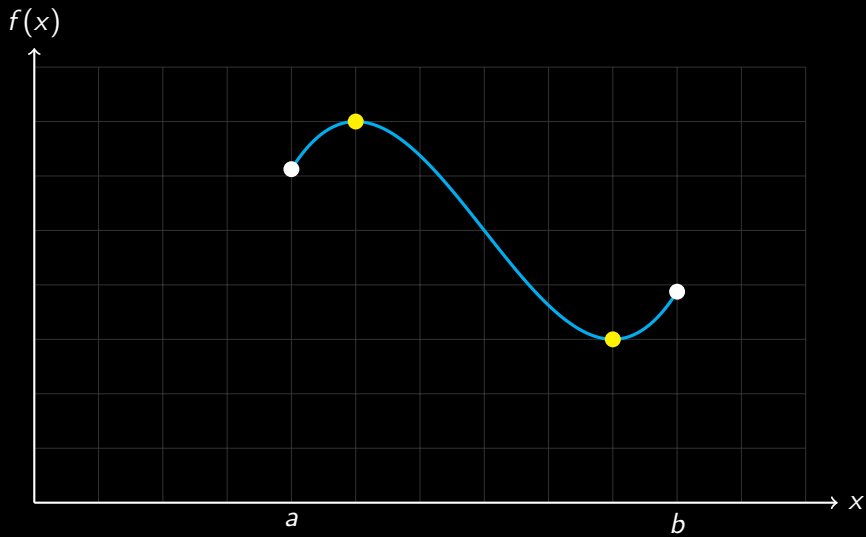
1. Find interior candidates where $f'(x) = 0$ or $f'(x)$ is undefined.
2. Evaluate $f(x)$ at every candidate and at both endpoints, a and b .
3. Compare the values: the largest is the absolute maximum and the smallest is the absolute minimum.

Always check both endpoints.









Optimisation

A standard optimisation process is:

1. Define the quantity to maximise or minimise: the **objective**.
2. Write a **constraint** connecting the variables.
3. Express the objective using one variable.
4. Differentiate and solve for feasible stationary points.
5. Classify the result and check relevant endpoints.
6. State the answer in context, including units.

Optimisation: Maximum Rectangle Area

A rectangle has perimeter 40 m. Find the dimensions that maximise its area.

$$\begin{aligned}2x + 2y &= 40, & y &= 20 - x, \\ A &= xy, & A(x) &= x(20 - x) = 20x - x^2.\end{aligned}$$

$$\begin{aligned}A'(x) &= 20 - 2x = 0, & x &= 10, \\ A''(x) &= -2 < 0.\end{aligned}$$

Therefore the area is maximised when $x = 10$ m and $y = 10$ m.

$$A_{\max} = 100 \text{ m}^2.$$

Optimisation: Business Revenue

A company's monthly revenue after spending x dollars on advertising, where $x \geq 0$, is modelled by

$$R(x) = 80\,000 + 50xe^{-x/4000}.$$

How much should the company spend to maximise its monthly revenue?

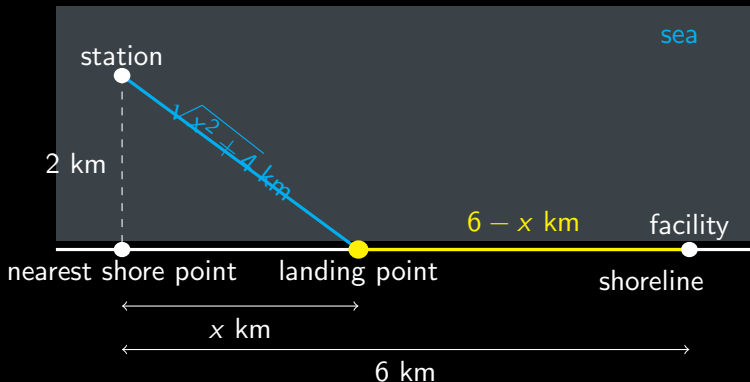
$$\begin{aligned} R'(x) &= 50e^{-x/4000} - \frac{50x}{4000}e^{-x/4000} \\ &= 50e^{-x/4000} \left(1 - \frac{x}{4000}\right). \end{aligned}$$

Since $e^{-x/4000} > 0$, $R'(x) = 0$ when $x = 4000$. The derivative changes from positive to negative at this value.

The company should spend \$4000 on advertising.

Optimisation: Underwater Cable

A monitoring station is 2 km offshore. A processing facility is 6 km along the shoreline from the nearest point on shore. Underwater cable costs \$5000/km and land cable costs \$3000/km. Where should the cable reach the shore to minimise the total cost?



Underwater Cable: Cost Function

Let x be the distance along the shore from the nearest shore point to the cable landing point.

$$\begin{aligned}\text{underwater length} &= \sqrt{x^2 + 2^2} = \sqrt{x^2 + 4}, \\ \text{land length} &= 6 - x.\end{aligned}$$

The total cost is therefore

$$\begin{aligned}C(x) &= 5000\sqrt{x^2 + 4} + 3000(6 - x), \\ 0 &\leq x \leq 6.\end{aligned}$$

The geometry provides the constraint: increasing the underwater distance decreases the distance travelled on land.

Underwater Cable: Minimum Cost

Differentiate and solve $C'(x) = 0$:

$$C'(x) = \frac{5000x}{\sqrt{x^2 + 4}} - 3000 = 0,$$

$$\frac{5x}{\sqrt{x^2 + 4}} = 3,$$

$$25x^2 = 9(x^2 + 4),$$

$$16x^2 = 36, \quad x = 1.5 \text{ km.}$$

Since

$$C''(x) = \frac{20\,000}{(x^2 + 4)^{3/2}} > 0,$$

this gives a minimum. The underwater length is 2.5 km, the land length is 4.5 km, and the minimum total cost is \$26 000.

Small Change Formula

If x changes by a small amount Δx , then the corresponding change in $y = f(x)$ is approximately

$$\Delta y \approx f'(x)\Delta x.$$

This is the tangent-line approximation near the known input x .

Small Changes: A Laboratory Model

In a laboratory experiment, the measured response P is modelled by

$$P(x) = 3 \sinh(2x - 6).$$

Develop a formula to estimate the change in P when x is near 3.

$$P'(x) = 6 \cosh(2x - 6),$$

$$P'(3) = 6 \cosh(0) = 6.$$

Let $\Delta x = x - 3$. Using the small-change formula,

$$\begin{aligned}\Delta P &\approx P'(3)\Delta x \\ &= 6\Delta x = 6(x - 3).\end{aligned}$$

Since $P(3) = 0$, this also gives $P(x) \approx 6(x - 3)$ near $x = 3$.

Small Changes: Checking the Approximation

Suppose the laboratory input changes from $x = 3$ to $x = 3.02$.
Then

$$\Delta x = 0.02.$$

The small-change formula gives

$$\begin{aligned}\Delta P &\approx 6\Delta x \\ &= 6(0.02) = 0.12.\end{aligned}$$

The exact change is

$$\begin{aligned}P(3.02) - P(3) &= 3 \sinh(0.04) - 0 \\ &\approx 0.120032.\end{aligned}$$

The absolute error is approximately 0.000032, so the linear estimate is very accurate for this small change in x .

Small Changes: Measurement Error

A sphere has measured radius 10 cm with possible error ± 0.02 cm.

$$V = \frac{4}{3}\pi r^3,$$
$$\Delta V \approx 4\pi r^2 \Delta r.$$

At $r = 10$ and $|\Delta r| = 0.02$,

$$\begin{aligned} |\Delta V| &\approx 4\pi(10)^2(0.02) \\ &= 8\pi \approx 25.1 \text{ cm}^3. \end{aligned}$$

Small Changes: Practice

A company's monthly profit, in dollars, is modelled by

$$P(x) = 800 \left(\frac{x}{800} + 5 \right)^{3/2},$$

where x is the number of products sold.

1. Find $P'(x)$.
2. Find the monthly profit when 3200 products are sold.
3. Use $\Delta P \approx P'(x)\Delta x$ to estimate the change in profit when sales increase from 3200 to 3280 products.

Small Changes: Answers

$$P'(x) = \frac{3}{2} \sqrt{\frac{x}{800} + 5},$$

$$P(3200) = 800(9)^{3/2} = \$21\,600.$$

Here $x = 3200$ and $\Delta x = 80$, so

$$\begin{aligned}\Delta P &\approx P'(3200)\Delta x \\ &= \frac{3}{2}\sqrt{9}(80) \\ &= 4.5(80) = \$360.\end{aligned}$$

The estimated increase in monthly profit is \$360.

Applications of Derivatives: Summary

- ▶ Solve $f'(x) = 0$ to locate stationary points.
- ▶ Classify using the first- or second-derivative test.
- ▶ Use $f''(x)$ to determine concavity.
- ▶ Confirm a sign change before declaring an inflection point.
- ▶ Optimise by combining an objective with a constraint.
- ▶ Approximate small changes using $\Delta y \approx f'(x)\Delta x$.